Sruti Bapatla

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EDUCATION

Carnegie Mellon University, B.S. ***September 2021 – May 2026***
Major: Mechanical Engineering
Additional Majors: Biomedical Engineering (Medical Device Track), Double Bass Performance
- **GPA**: 3.85/4.00, University Honors, Dean's List (All Semesters)

Veritas University (Study Abroad), San José, Costa Rica ***September 2024 – December 2024***
Focus: Environmental Science, Sustainable Development, Spanish

ENGINEERING EXPERIENCE

Neocis ***June 2025 – August 2025***
Surgical Instruments Intern
- Redesigned an extra-oral fiducial array for a dental surgery robot so patients can be oxygenated by mask instead of intubation during a hypoxemic event (~7% of procedures), while keeping the original strength and work clearance.
- Prototyped 20 iterations that passed stress, deflection, and mask-clearance tests, then refined form, comfort, and ease of understanding from surgeon feedback; the design advanced to DFM review for injection molding.
- Designed a gravity-fed hopper fixture, using SolidWorks and 3D printing, that drops all 12 fluorescent beads into fiducial pockets with one button press, replacing placement by hand and reducing manufacturing time. Observed operators to add a thumb rest and an audible completion click, and built the CAD parametrically to adapt to changing fiducial shapes.

Duke University ***September 2026 – Present***
R&D Engineer I
- Designing a semester-long electric car project platform in which sophomores build an analog circuit lab by lab, making design decisions such as choosing op-amps, transistors, and motors.
- Researching student use of generative AI in engineering courses and designing projects that strengthen design fundamentals.

Smith & Nephew ***June 2024 – August 2024***
Robotics - Hardware Engineering Intern
- Adapted self-tapping bone screws and implant fixtures from knee surgery for the smaller, more fragile bones of the shoulder.
- Benchmarked six internal and industry self-tapping screw patterns through SolidWorks simulation and sawbone compression and deflection testing.
- Designed custom self-tapping screws from the thread profile and pitch of the top three performers, and wrote the experimental design validation protocol.
- Tested the custom screws in cadaver labs and used the results to calibrate the simulation's bone material and boundary conditions; my design produced the lowest bone stress and deflection and moved to the product team for refinement before production.

Smith & Nephew ***June 2023 – August 2023***
Manufacturing Engineering Intern
- Programmed 135 knee and hip zirconium oxide implants in VEI software to validate part size and family during manufacturing and packaging process.
- Created fixtures for implants during scanning, plus VRA documentation and work instructions, and trained operators on the new system in the clean room.

Caterpillar ***June 2022 – August 2022***
R&D Engineering Intern
- Automated and ran daily tests on a dual-fuel engine to check engine health and diagnose turbine efficiency issues, including standard test cell safety and quality checks.
- Created NI DIAdem scripts to analyze and visualize steady-state and transient engine data to better advertise engines to customers.

Government of Republic of Palau ***May 2026 – July 2026***

Technology Consultant Intern

- Interviewed ministry staff and rebuilt a layered HR record system in Microsoft Access so data entered once carries through every connected system, reducing the work of three people to one.
- Built Power BI dashboards that automate report generation, iterating on colors and information density to match Palauan communication preferences.
- Added drag-and-drop modules that staff can enable without technical support, based on interviews about future needs, and led trainings with picture-based documentation to overcome language barriers.
- Developed an AI chatbot running on a local model for the ministry website.

NOTABLE PROJECT WORK

Spice Container with Integrated Measuring System ***September 2025 – December 2025***

- Designed the lead-screw and belt-and-pulley dispensing mechanism for a spice jar sized container that measures ¼ tsp to 1 tbsp in a single press.
- Toleranced the assembly to fit a 1.8" × 3.2" × 4" form factor, shrinking the design across multiple prototype iterations to fit in a standard spice rack.

Spinal Torque Measuring Adapter (Medtronic-Sponsored Capstone Project) ***September 2023 – May 2024***

- Designed and manufactured a biocompatible adapter for measuring applied torque on spinal surgical instruments.
- Integrated a load cell and strain gauge with an Arduino to measure up to 11 N·m of torque within ±0.25 N·m, calibrated against a torque wrench per ISO 6789:2017.
- Built a computer interface with live and peak torque readings, tare, and color feedback for “safe” torque, refined through feedback from surgeons and Medtronic advisors.

Accessible Power Drill Adapter ***September 2023 – December 2023***

- Designed a drill attachment for users with tendonitis that replaces the trigger grip with a palm-actuated rear handle and wrist brace, based on user surveys.
- Led stress analysis and CAD prototyping in SolidWorks, and validated the handle with FEA and hand calculations.

Music Box ***January 2025 – May 2025***

- Built a working music box from scratch, tuning each comb tine with cantilever-beam vibration theory across several comb prototypes and mapping a song (baby elephant walk) onto the drum's pin layout.

Music MATLAB ***January 2024 – May 2024***

- Built a MATLAB pipeline that converts audio recordings into sheet music, using a fast Fourier transform to identify the dominant note in each 0.05 s slice.

SkinVive: Facial Burn Scar Covering ***January 2025 – May 2025***

- Researched the FDA regulatory pathway for a layered silicone–hydrogel scar covering, classifying it as a Class I, 510(k)-exempt device and identifying the ISO 10993 biocompatibility testing needed before clinical use.

OUTREACH & LEADERSHIP EXPERIENCE

Carnegie Mellon Fifth Year Scholar ***September 2025 – May 2026***

- One of three students awarded a year-long fellowship to develop a project with strong community impact.
- Launched a nonprofit-sponsored mechanical engineering capstone program, shaped with feedback from six local nonprofit groups to identify gaps in local community nonprofits that could be addressed by mechanical engineering students.
- Developed the course curriculum, coordinated with the teaching professor, and created a financial plan for the program's sustainability.

Carnegie Mellon Mechanical Engineering Student Advisory Council Leader ***September 2022 – May 2025***

- Brought feedback from mechanical engineering students to faculty to improve the student experience.

Carnegie Mellon Lead Campus & Mechanical Engineering Tour Guide ***January 2022 – May 2026***

- Gave prospective students and families a personalized glimpse into campus life, and helped train new tour guides.

Society of Women Engineers, Carnegie Mellon University Chapter, Advocacy Chair ***September 2021 – September 2022***

- Organized advocacy events, such as voter registration drives and DEI programming.

Carnegie Mellon Makerspace Student Worker ***January 2024 – May 2026***

- Helped students fabricate projects on the waterjet, 3D printers, and laser cutters.

Bull City Woodshop, Volunteer Woodshop Instructor ***September 2026 – Present***
- Teaching adult woodshop classes to build benches placed at bus stops across Durham.

Durham Bike Co-op, Volunteer Bike Mechanic ***September 2026 – Present***
- Repairing donated bikes and performing tune-ups on community members' bikes.

Carroll Street Park, Volunteer Team Lead ***September 2026 – Present***
- Helping turn a local park into a community space: removing invasive species, researching native plants, leading volunteer teams, partnering with local organizations, and designing open-ended play toys for a range of ages.

Literacy Pittsburgh, English Tutor ***August 2023 – Present***
- Tutoring a non-native English-speaking adult three times a week with personalized lesson plans built around the student's learning goals.

TEACHING EXPERIENCE

Duke | First Year Design Technical Advisor ***September 2026 – Present***
- Providing design feedback regarding stakeholder needs, design methodology, and fabrication methods.

Carnegie Mellon | DIY Design and Fabrication Teaching Assistant ***January 2026 – May 2026***
- Directed lab sessions on silicone molding, 3D printing, laser cutting, and foam modeling.

Carnegie Mellon | Mechanical Systems Experimentation Teaching Assistant ***September 2025 – May 2026***
- Developed 1DOF and 2DOF torsional and rectilinear system experiments in Simulink that prioritize students' decision-making during lab.

Carnegie Mellon | Mechanical Design: Methods & Applications Teaching Assistant ***September 2025 – December 2025***
- Restructured product adaptation for individuals with disabilities project so students set quantitative requirements from stakeholder research.
- Led team of four undergraduate student teaching assistants to ensure fair and consistent grading.

Carnegie Mellon | Mechanics II: 3D Design Teaching Assistant ***January 2024 – May 2024, January 2025 – May 2025***
- Developed materials list to align with students' fabrication capabilities for chair building project, focused on manual machining, 3D printing, and laser cutting.
- Led CAD and fabrication training sessions, and weekly office hours to groups of ~5 students.

Carnegie Mellon | Mechanics I: 2D Design Teaching Assistant ***January 2026 – May 2026***
- Developed lecture and project materials for a project in which students design and build their own four-bar linkage to trace a specified trajectory.

Carnegie Mellon | Project Fabrication & Assembly Teaching Assistant ***September 2025 – May 2026***
- Taught graduate students the safe operation of tools including drills, oscillating tools, jigsaws, hole saws, and more.

SKILLS

- **Software**: Python, MATLAB, SolidWorks, ANSYS, Onshape, Power BI, Simulink, Fusion 360, Java, Microsoft Access
- **Fabrication**: Mill, Lathe, CNC, Water Jet, Laser Cutter, 3D Printer, Soldering, PCB Design
- **Languages:** Spanish (proficient in reading, writing, and speaking), Telugu (native proficiency)

AWARDS

- Andrew Carnegie Scholar | 1 of 40 students selected as a senior representative for academic excellence, leadership, and service
- Carnegie Mellon Fifth Year Scholar | One of three students selected to spend a year on a community development project at CMU
- Carnegie Mellon Senior Leadership Recognition | Selected for making significant contributions to CMU community
- Carnegie Mellon Mechanical Engineering Undergraduate Service and Leadership Award
- Mortar Board Honors Society
- Pi Tau Sigma Honors Society (President)